VINEETHA KOLLAMARAM

Bengaluru, India | kollamoramvineetha@gmail.com| https://www.linkedin.com/in/vineetha-kollamoram/ |
https://github.com/kollamoramvineetha

PROFESSIONAL SUMMARY

QA Automation Engineer with 2+ years of experience in Python-based test automation, including UI and API testing, SQL-driven test reporting, and AI response validation. Skilled in building automated test frameworks across Android and iOS platforms (Kotlin, Swift), validating functional and AI system behavior, and evaluating agentic AI systems including tool/function calling and multi-step agent workflows. Currently expanding this expertise through an ongoing course in Agentic AI. Improved application stability to above 90%. Seeking an Automation Engineer or AI/Agentic AI Testing role to apply automation testing expertise to build robust, scalable test suites for both traditional and AI-driven systems.

CORE SKILLS

- **Testing & QA:** Automation Testing, Functional Testing, Regression Testing, End-to-End (E2E) Testing, STLC, EarlGrey Framework, Test Case Design, API Testing, UI Testing, Defect Resolution, Root Cause Analysis
- **Agentic AI & Function Calling:** Function/Tool Calling Concepts, Agent Loop & Orchestration, Parallel & Compositional Tool Calling, Model Context Protocol (MCP), Tool Description Design, Agent Failure-Mode Testing (Gemini API)
- **AI & LLM Technologies**: LLM Evaluation, Prompt Engineering, Prompt Optimization, AI Response Validation
- **Programming Languages:** Python, Java, C
- **Frontend Development:** HTML, CSS, Bootstrap, JavaScript, React.js
- **Databases**: MySQL, SQL (Queries, Joins, Views)
- **Cloud & Tools:** AWS, Agile/Scrum, Version Control, Dashboard Development

PROFESSIONAL EXPERIENCE

Software Engineer 2 | *Raja Software Labs*April 2024 – Present

Automation Testing, AI Response Validation & Prompt Optimization

AI Model Testing (Gemini Assistant)

- Designed comprehensive test scenarios, test cases, and test data to validate Gemini Assistant AI responses, managing defects throughout the Software Testing Life Cycle (STLC).
- Validated Gemini Assistant responses across diverse query types — media/entertainment commands, general web-based questions, and smart-device actions — ensuring correct, consistent AI behavior across UI and API layers.
- Automated AI response validation using the Inception framework (Python) and Python-based test frameworks, strengthening functional and regression testing coverage.
- Evaluated AI assistant behavior in tool/function-call interactions, applying knowledge of agentic AI patterns (tool selection accuracy, parameter extraction, multi-step reasoning) to strengthen test coverage.
- Created and refined prompts to improve AI response quality, contextual understanding, and performance; identified inaccuracies, prompt gaps, and logical/system defects, fixing several directly.
- Conducted root cause analysis, debugging, retesting, and defect resolution for AI workflows, collaborating on internal code fixes to improve model precision and reliability.
- Improved application stability to above 90% by increasing test reliability, reducing flaky test cases, and strengthening QA processes.
- Collaborated with development teams to improve software performance, AI system effectiveness, and release confidence.

API Testing (Google Home App — Device Integrations)

- Performed API testing for various connected devices integrated with the Google Home app, validating device-to-API interactions across smart-home use cases.
- Identified and fixed numerous device API issues, applying SDK knowledge to troubleshoot and report product-level defects; also resolved several infrastructure-related issues.
- Wrote new automated test methods in Kotlin and Swift, executing API tests across both Android and iOS platforms.
- Built strong working knowledge of the Bazel build system to support efficient build, test, and validation workflows.
- Worked with the EarlGrey framework for UI and integration testing.

Dashboard Development (SQL)

- Designed and developed an end-to-end test results dashboard using SQL on a MySQL database, displaying the complete historical view of test case execution over time.
- Wrote optimized SQL queries, joins, and views to generate key metrics such as total test case count, automated test count, blocked test count, execution trends, and overall stability/pass rate percentage.
- Enabled clients to directly view test case counts and review overall automation coverage and pass rate without manual reporting, significantly improving visibility into testing progress.
- Displayed failure reasons directly on the dashboard, enabling faster triage and stabilization of failing test cases by allowing team members to quickly identify and address errors.
- Provided data insights that helped track and maintain product quality consistency.

Leadership & Collaboration

- Played a leadership role by coordinating with team members and ensuring smooth execution of tasks.
- Managed and mentored a team of 4+ members, providing guidance, task allocation, and support to ensure successful project delivery.
- Conducted training sessions for junior team members and assigned learning tasks to support their technical growth and skill development.
- Tracked team progress regularly, provided constructive feedback, and helped resolve technical challenges.
- Participated in the hiring process by screening candidates, reviewing profiles, and providing feedback to support recruitment decisions.
- Assisted in onboarding new team members and helping them understand project workflows, tools, and processes.
- Managed multiple assignments simultaneously and ensured timely completion.
- Provided weekly progress tracking updates to stakeholders and team members.
- Identified risks early and ensured issues were resolved efficiently.
- Collaborated with cross-functional teams following Agile practices.

PROJECTS

Guess My Number

- Built an interactive number-guessing game in React.js where users guess a system-generated number within 5 attempts.

E-Commerce Website

- Developed an online shopping platform allowing users to browse a product catalog and manage items in a shopping cart.

Front-End Websites

- Designed and built multiple responsive front-end websites (Simplotel, Motherson Project) using HTML, CSS, Bootstrap, and JavaScript.

EDUCATION

B.Tech, Computer Science Engineering — CGPA: 8.87
Srinivasa Ramanujan Institute of Technology, Anantapur 2019 – 2023

Intermediate (MPC) — CGPA: 9.88
Sri Chaitanya Junior College, Vijayawada 2017 – 2019

Secondary School Certificate — CGPA: 9.7
Sri Chaitanya School, Dharmavaram 2016 – 2017

CERTIFICATIONS

- Agentic AI (Frontlines Media) — in progress, started August 2026
- AWS Certified Cloud Practitioner (Global)
- Programming with Python
- Basics of SQL